$$= 2 + [x - 2\log x]_2^x = 2 + x - 2\log x - 2 + 2\log 2$$

$$= x - 2\log x + 2\log 2$$

2nd approximation

$$y_2 = y_0 + \int_{x_0}^{x} f(x, y_1)\,dx = 2 + \int_2^x f(x, x - 2\log x + 2\log 2)\,dx$$

$$= 2 + \int_2^x \left(1 - \frac{x - 2\log x + 2\log 2}{x}\right) dx$$

$$= 2 + \int_2^x \left(1 - 1 + \frac{2\log x}{x} - \frac{2\log 2}{x}\right) dx$$

$$= 2 + \left[2 \cdot \frac{1}{2}\{\log (x)\}^2 - 2\log 2 \log x\right]_2^x$$

$$= 2 + \{\log(x)\}^2 - 2\log 2 \log x - (\log 2)^2 + 2(\log 2)^2$$

$$= 2 + (\log x)^2 - 2\log 2 \log x + (\log 2)^2$$

$$\therefore y(x) = 2 + (\log x)^2 - 2\log 2 \log x + (\log 2)^2$$

8.A.6. Euler's method :[NUH–04, 06, 08(Old),08,13, DUH–06]

We have so farther yield the solution of a differential equation in the form of a power series. We will now proceed to describe methods which give the solution in the form of a set of tabulated values at equally spaced points.

We consider the differential equation

$$\frac{dy}{dx} = f(x, y) \cdots\cdots (1), \text{ with } y(x_0) = y_0 \cdots\cdots (2)$$

Suppose that we want to solve the equation (1) for y at

$x = x_r = x_0 + rh,\ r = 1, 2, 3, \cdots$

Integrating (1) we obtain

$$\int_{y_0}^{y_1} dy = \int_{x_0}^{x_1} f(x, y)\,dx \Rightarrow [y]_{y_0}^{y_1} = \int_{x_0}^{x_1} f(x, y)\,dx$$

$$\therefore y_1 = y_0 + \int_{x_0}^{x_1} f(x, y)\,dx \cdots\cdots (3)$$

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Assuming that $f(x, y) = f(x_0, y_0)$ in $x_0 \le x \le x_1$, we get

$$y_1 = y_0 + (x_1 - x_0) f(x_0, y_0) \quad \therefore y_1 = y_0 + hf(x_0, y_0) \text{ ..... (4)}$$

Similarly for the range $x_1 \le x \le x_2$, we have

$$y_2 = y_1 + \int_{x_1}^{x_2} f(x, y)\, dx$$

Assuming that $f(x, y) = f(x_1, y_1)$ in $x_1 \le x \le x_2$, we obtain

$$y_2 = y_1 + hf(x_1, y_1)$$

Proceeding in this way, we get the general formula

$$y_{n+1} = y_n + hf(x_n, y_n),\ n = 0, 1, 2, 3, \ldots$$

This formula is known as Euler's formula.

This process is very slow. For practical use, the method is unsuitable because to get reasonable accuracy with this method we need to take a smaller value for h.

Example–22. Solve by Euler's method of the equation $\frac{dy}{dx} = x + y$, $y(0) = 1$ at the point $x = 0.05$ and $x = 0.10$ taking $h = 0.05$.

[NUH–1998, 2008(Old), 2013]

Solution : The given differential equation is

$$\frac{dy}{dx} = x + y, \quad \therefore f(x, y) = x + y$$

Also we have $x_0 = 0$, $y_0 = 1$ and $h = 0.05$

We know that the Euler's formula for the numerical solution of the differential equation.

$$\frac{dy}{dx} = f(x, y) \text{ is } y_{n+1} = y_n + hf(x_n, y_n) \text{ ..... (1)}$$

Putting $n = 0$ in (1), we get

$$y_1 = y_0 + hf(x_0, y_0) = y_0 + h(x_0 + y_0) = 1 + 0.05(0 + 1) = 1.05$$

$\therefore y(0.05) = 1.05$

Now putting $n = 1$ in (1), we get

$$y_2 = y_1 + hf(x_1, y_1)$$

$$\Rightarrow y_2 = y_1 + h(x_1 + y_1)$$

$$= 1.05 + .05(.05 + 1.05)$$

$$= 1.1050$$

$\therefore y(0.1) = 1.1050$

$\because x_1 = x_0 + h$

$= 0 + .05$

$= .05$

Example–23. Using Euler's method solve $\frac{dy}{dx} = 1 + xy$ with $y(0) = 2$. Find $y(0.1)$, $y(0.2)$ and $y(0.3)$.

Solution : The given differential equation is

$$\frac{dy}{dx} = 1 + xy, \quad \therefore f(x, y) = 1 + xy$$

Also we have $x_0 = 0$, $y_0 = 2$ and $h = 0.1$

we know that the Euler's formula for the numerical solution of the differential equation

$$\frac{dy}{dx} = f(x, y) \text{ is } y_{n+1} = y_n + hf(x_n, y_n) \ldots\ldots (1)$$

Putting $n = 0$ in (1), we get $y_1 = y_0 + hf(x_0, y_0)$

$$\Rightarrow y_1 = y_0 + h(1 + x_0y_0) = 2 + 0.1(1 + 0) = 2.1$$

$\therefore y(0.1) = y_1 = 2.1$

Now $x_1 = x_0 + h = 0 + 0.1 = 0.1$

Putting $n = 1$ in (1) we get

$$y_2 = y_1 + hf(x_1, y_1) = y_1 + h(1 + x_1y_1) = 2.1 + 0.1(1 + 0.1 \times 2.1) = 2.221$$

$\therefore y(0.2) = y_2 = 2.221$

Again, $x_2 = x_1 + h = 0.1 + 0.1 = 0.2$

Putting $n = 2$ in (1), we get

$$y_3 = y_2 + hf(x_2, y_2) = 2.221 + 0.1 \times (1 + 0.2 \times 2.221) = 2.36542$$

$\therefore y(0.3) = y_3 = 2.36542.$

Example–24. Use Euler's method with $h = 0.1$ to find the solution of $\frac{dy}{dx} = x^2 + y^2$, $y(0) = 0$ in the range $0 \le x \le 0.5$.

Solution : The given differential equation is $\frac{dy}{dx} = x^2 + y^2$

$\therefore f(x, y) = x^2 + y^2$

Also we have $x_0 = 0$, $y_0 = 0$, $h = 0.1$

Using $x_r = x_0 + rh$, $r = 1, 2, 3, 4, 5$, we get

$x_1 = x_0 + h = 0 + 0.1 = 0.1$

$x_2 = x_0 + 2h = 0 + 0.2 = 0.2$

$x_3 = 0.3$, $x_4 = 0.4$, $x = 0.5$

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We know that the Euler's formula for the numerical solution of the differential equation $\frac{dy}{dx} = f(x, y)$ is

$$y_{n+1} = y_n + hf(x_n, y_n) \ldots\ldots (1)$$

Putting n = 0 in (1), we get

$$y_1 = y_0 + hf(x_0, y_0) = y_0 + h(x_0^2 + y_0^2) \quad [\because f(x, y) = x^2 + y^2]$$

$\Rightarrow y_1 = 0 + 0.1(0 + 0) = 0, \quad \therefore y(0.1) = 0$

Putting n = 0 in (1), we get

$$y_1 = y_0 + hf(x_0, y_0) = y_0 + h(x_0^2 + y_0^2) \quad [\because f(x, y) = x^2 + y^2]$$

$\Rightarrow y_1 = 0 + 0.1(0 + 0) = 0, \quad \therefore y(0.1) = 0$

Putting n = 1 in (1), we get

$y_2 = y_1 + hf(x_1, y_1)$ $\quad [\because f(x, y) = x^2 + y^2$

$= 0 + 0.1 \times (0.1^2 + 0) = 0.001$ $\quad \therefore f(x_1, y_1) = x_1^2 + y_1^2]$

$\therefore y(0.2) = 0.001$

Again, Putting n = 2 in (1), we get

$y_3 = y_2 + hf(x_2, y_2)$ $\quad [\because f(x, y) = x^2 + y^2$

$= 0.001 + 0.1(0.2^2 + 0.001^2)$ $\quad f(x_2, y_2) = x_2^2 + y_2^2]$

$= 0.005 \quad \therefore y(0.3) = 0.005$

Putting n = 3 in (1), we get

$y_4 = y_3 + hf(x_3, y_3) \Rightarrow y_4 = 0.005 + 0.1(0.3^2 + 0.005^2)$

$\therefore y_4 = 0.014 \quad \therefore y(0.4) = y_4 = 0.014$

Putting n = 4 in (1), we get

$y_5 = y_4 + hf(x_4, y_4) = 0.014 + 0.1\ (0.4^2 + 0.014^2) = 0.03002$

$\therefore y(0.5) = 0.03002.$

Example–25. Solve $\frac{dy}{dx} = 1 - y$, $y(0) = 0$ using Euler's method. Find y at x = 0.1 and x = 0.2. Compare the result with results of exact solution.

Solution : The given differential equation is $\frac{dy}{dx} = 1 - y$

$\therefore f(x, y) = 1 - y$

Also we have $x_0 = 0$, $y_0 = 0$ and $h = 0.1$

we know that the Euler's formula for the numetical solution of the differential equation $\frac{dy}{dx} = f(x, y)$ is

$$y_{n+1} = y_n + hf(x_n, y_n) \text{ ..... (1)}$$

Putting n = 0 in (1), we get

$y_1 = y(0.1) = y_0 + hf(x_0, y_0)$

$= 0 + 0.1 \times (1 - 0)$ [$\because f(x, y) = 1 - y$

$= 0.1$ $\therefore f(x_0, y_0) = 1 - y_0$]

Now $x_1 = x_0 + h = 0 + 0.1 = 0.1$

Putting n = 1 in (1), we get

$y_2 = y_1 + hf(x_1, y_1) \Rightarrow y(0.2) = 0.1 + 0,1 \times (1 - 0.1) = 0.19$

Hence y(0.1) = 0.1 and y(0.2) = 0.19

Again, $\frac{dy}{dx} = 1 - y \Rightarrow \frac{dy}{1 - y} = dx$

$\therefore \ln(1 - y) = -x + c$ ..... (2) [by integrating]

Using initial condition y(0) = 0 in (2), we get

$\ln 1 = 0 + c$ $\therefore c = 0$ [$\because \ln 1 = 0$]

$\therefore$ (2) becomes $\ln(1 - y) = -x \Rightarrow 1 - y = e^{-x}$

$\therefore y = 1 - e^{-x}$ $\therefore y(0.1) = 1 - e^{-0.1} = 0.1052$

and $y(0.2) = 1 - e^{-0.2} = 0.2214$.

Example–26. Using Euler's method solve $\frac{dy}{dx} = -y$ with y(0) = 1. Find y(0.01), y(0.02), y(0.03), y(0.04). Also compare the result with results of exact solution.

Solution : The given differential equation is

$$\frac{dy}{dx} = -y \quad \therefore f(x, y) = -y$$

Also we have $x_0 = 0$, $y_0 = 1$ and h = 0.1

Here $x_1 = 0.01$, $x_2 = .02$, $x_3 = 0.03$, $x_4 = 0.04$ and h = 0.01

we have to find y_1, y_2, y_3 and y_4

We know that the Euler's formula for the numerical solution of the differenlial equation $\frac{dy}{dx} = f(x, y)$ is

$$y_{n+1} = y_n + hf(x_n, y_n) \text{ ..... (1)}$$

Putting n = 0 in (1), we get

$y_1 = y_0 + hf(x_0, y_0) \Rightarrow y_1 = y_0 + h(-y_0) = 1 + 0.01\,(-1) = 0.99$

$\therefore y(0.01) = y_1 = 0.99$

Now $x_1 = x_0 + h = 0 + 0.01 = 0.01$

Putting n = 1 in (1), we get

$y_2 = y_1 + hf(x_1, y_1) = 0.99 + 0.01 \times (-0.99) = 0.9801$

$\therefore y(0.02) = y_2 = 0.9801$

Again, $x_2 = x_1 + h = 0.01 + 0.01 = 0.02$

Putting n = 2 in (1), we get

$y_3 = y_2 + hf(x_2, y_2)$ $\quad \because f(x_2, y_2) = -y_2$

$= 0.9801 + 0.01 \times (-0.9801) = 0.9703.$

Similarly $y_4 = y_3 + hf(x_3, y_3)$

$= 0.9703 + 0.01 \times (-0.9703)$

$= 0.9606$

Again, $\frac{dy}{dx} = -y \Rightarrow \frac{dy}{y} = -dx$

$\therefore \ln y = -x + c$ ..... (2) [by integrating]

Using initial condition y(0) = 1 in (2), we get

$\ln 1 = -0 + c \quad \therefore c = 0$ [$\because \ln 1 = 0$]

$\therefore$ (2) becomes, $\ln y = -x \quad \therefore y = e^{-x}$

$\therefore$ The exact value of y at x = 0.01, 0.02, 0.03 and 0.04

$\therefore y(0.01) = 0.99$, $y(0.02) = 0.9802$, $y(0.03) = 0.9704$ and $y(0.04) = 0.0608$

Example–27. Solve $\frac{dy}{dx} = \frac{y - x}{y + x}$, with y(0) = 1. Find y approximately for x = 0.1 by taking h = .02 **[NUH–2002 (Old)]**

Solution : The given differential equation is

$$\frac{dy}{dx} = \frac{y - x}{y + x} \qquad \therefore f(x, y) = \frac{y - x}{y + x}$$

Also we have $x_0 = 0$, $y_0 = 1$ and h = 0.02

Here $x_1 = .02$, $x_2 = .04$, $x_3 = .06$, $x_4 = 0.8$, $x_5 = .1$

We know that the Euler's formula for the numerical solution of the differential equation $\frac{dy}{dx} = f(x, y)$ is

$$y_{n+1} = y_n + hf(x_n, y_n) \quad \ldots\ldots (1)$$

Now we shall find the values of y_1, y_2, y_3, y_4 and y_5. For this purpose putting n = 0, 1, 2, 3, 4 in (1) respectively, we get

$$y_1 = y_0 + hf(x_0, y_0) = y_0 + h\frac{y_0 - x_0}{y_0 + x_0} = 1 + 0.02\left(\frac{1-0}{1+0}\right) = 1.02$$

$$y_2 = y_1 + hf(x_1, y_1) = y_1 + h\frac{y_1 - x_1}{y_1 + x_1}$$

$$= 1.02 + (0.02)\left(\frac{1.02 - .02}{1 + .02}\right) = 1.0392$$

$$y_3 = y_2 + hf(x_2, y_2) = y_2 + h\frac{y_2 - x_2}{y_2 + x_2}$$

$$= 1.0392 + 0.02 \times \frac{1.0392 - .04}{1.0392 + .04} = 1.0577$$

$$y_4 = y_3 + hf(x_3, y_3) = y_3 + h\frac{y_3 - x_3}{y_3 + x_3}$$

$$= 1.0577 + 0.02 \times \frac{1.0577 - .06}{1.0577 + .06} = 1.0756$$

$$y_5 = y_4 + hf(x_4, y_4) = y_4 + h\frac{y_4 - x_4}{y_4 + x_4}$$

$$= 1.0756 + 0.02 \times \frac{1.0756 - .08}{1.0756 + .08} = 1.0928$$

Hence $y(0.1) = y_5 = 1.0928$.

Example–28. Using Euler's method, solve the differential equation $\frac{dy}{dx} - 1 = y^2$, $y(0) = 0$ tanking $h = 0.1$ and obtain $y(0.1)$, $y(0.2)$ and $y(0, 3)$. **[NUH–2005, 2014]**

Solution : $\frac{dy}{dx} - 1 = y^2 \Rightarrow \frac{dy}{dx} = 1 + y^2$, $\therefore f(x, y) = 1 + y^2$

Here $x_0 = 0$, $y_0 = 0$ and $h = 0.1 \Rightarrow x_1 = 0, 1, x_2 = 0.2, x_3 = 0.3$

$\therefore y_1 = y_0 + hf(x_0, y_0) = 0 + 0.1(1 + 0^2) = .1$

$\therefore y(0.1) = y_1 = 0.1$

$y_2 = y_1 + hf(x_1, y_1) = 0.1 + 0.1(1 + (0.1)^2) = 0.201$

$\therefore y(0.2) = y_2 = 0.201$

$y_3 = y_2 + hf(x_2, y_2) = 0.201 + 0.1(1 + (0.201)^2) = 0.305$

$\therefore y(0.3) = y_3 = 0.305$

$\therefore y(0.1) = 0.1,\ y(0.2) = 0.201,\ y(0.3) = 0.305$

8.A.7. Modified Euler's method :

[NUH–2004, 2008, 2012, 2013, DUH–2006]

By Euler's method, we have $y_1 = y_0 + hf(x_0, y_0)$ ..... (1)

Instead of approximating $f(x, y)$ by $f(x_0, y_0)$ in (3), we approx-ate it by $\frac{1}{2}[f(x_0, y_0) + f(x_1, y_1)]$, which is average of the slopes of e tangents at the points corresponding to $x = x_0$ and $x = x_1$.

Thus from (1) we obtain $y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1)]$

Here $y_1^{(1)}$ is the first modified value of y_1.

Let $y_1^{(2)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(1)})$

Proceeding in this way, we thus obtain the iteration formula

$y_1^{(n+1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(n)})],\ n = 0, 1, 2, \cdots$

where $y_1^{(n)}$ is the nth approximation to y_1.

8.A.8. Which method gives more acurate results and hy?

[NUH–2004, 2013]

Euler modify method gives more accurate results in the mple Euler method, we use the slope at the beginning of the terval, $\frac{dy}{dx} = y' = f(x, y)$, to determine the increment to the nction. This technique would be correct only if the function ere linear. What we need instead is the correct average slope th in the interval. In the modified Euler method this can be pproximated by the mean of the slopes at both ends of the terval.

For this reason Euler modify method gives more accurate sults.

Example–29. Find y for x = 0.2 and x = 0.5 from the ini value problem $\frac{dy}{dx} = ln\,(x + y)$, y(0) = 1 by Euler's modified meth [পরিবর্তিত অয়লার পদ্ধতিতে আদিমান সমস্যা থেকে x = 0.2 এবং x = 0.5 এর জ ের মান নির্ণয় কর।] [NUH–2007(0

Solution : Given, $\frac{dy}{dx} = ln\,(x + y)$

Here, $x_0 = 0$, $y_0 = 1$ and $h = 0.1$

By Euler's method we get, $y_1 = y_0 + hf(x_0, y_0)$

$= 1 + 0.1(0) = 1$

By Euler's modified method, we get

$$y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(0)})] = 1 + 0.05\,[0 + 0.0953] = 1.0047$$

$$y_1^{(2)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(1)})] = 1.00498$$

$$\therefore y_1^{(3)} = y + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(2)})] = 1.00498$$

$\therefore y(0.1) = 1.00499$

$$y_2 = y_1 + hf(x_1, y_1) = 1.01497$$

$$y_2^{(1)} = y_1 + \frac{h}{2}[f(x_1, y_1) + f(x_2, y_2^{(2)})]$$

$$= 1.00499 + 0.05[0.0998 + 0.19472] = 1.01972$$

$$y_2^{(2)} = y_1 + \frac{h}{2}[f(x_1, y_1) + f(x_2, y_2^{(1)})] = 1.01991$$

$$y_2^{(3)} = y_1 + \frac{h}{2}[f(x_1, y_1) + f(x_2, y_2^{(2)})] = 1.01991$$

$\therefore y(0.2) = 1.0199$

$$y_3 = y_2 + hf(x_2, y_2) = 1.0199 + 0.1(0.198769) = 1.039776$$

$$y_3^{(1)} = y_2 + \frac{h}{2}[f(x_2, y_2) + f(x_3, y_3^{(0)})] = 1.04446$$

$$y_3^{(2)} = y_2 + \frac{h}{2}[f(x_2, y_2) + f(x_3, y_3^{(1)})] = 1.044638$$

$$y_3^{(3)} = y_2 + \frac{h}{2}[f(x_2, y_2) + f(x_3, y_3^{(2)})] = 1.04464$$

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$\therefore y(0.3) = 1.04464$

$y_4 = y_3 + hf(x_3, y_3) = 1.04464 + 0.1(0.296126) = 1.07425$

$y_4^{(1)} = y_3 + \frac{h}{2}[f(x_3, y_3) + f(x_4, y_4^{(0)})] = 1.07885$

$y_4^{(2)} = y_3 + \frac{h}{2}[fx_3, y_3) + f(x_4, y_4^{(1)})] = 1.07901$

$y_4^{(3)} = y_3 + \frac{h}{2}[f(x_3, y_3) + f(x_4, y_4^{(2)})] = 1.07901$

$\therefore y(0.4) = 1.07901$

$y_5 = y_4 + hf(x_4, y_4) = 1.07901 + 0.1(0.39137) = 1.11815$

$y_5^{(1)} = y_4 + \frac{h}{2}[f(x_4, y_4) + f(x_5, y_5^{(0)})] = 1.12264$

$y_5^{(2)} = y_4 + \frac{h}{2}[f(x_4, y_4) + f(x_5, y_5^{(1)})] = 1.12278$

$y_5^{(3)} = y_4 + \frac{h}{2}[f(x_4, y_4) + f(x_5, y_5^{(2)})] = 1.12278$

$\therefore y(0.5) = 1.12278$

Thus, $y(0.2) = 1.0199$, $y(0.5) = 1.12278$

Example–30. Using modified Euler's method solve $\frac{y}{x} = 1 + xy$ with $y(0) = 2$. Find $y(0.1)$, $y(0.2)$ and $y(0.3)$.

Solution : Given $\frac{dy}{dx} = 1 + xy$ with $y(0) = 2$

$\therefore f(x, y) = 1 + xy$, $x_0 = 0$, $y_0 = 2$, $h = 0.1$

By Euler's formula, we get

$y_1 = y_0 + hf(x_0, y_0) = y_0 + h[1 + x_0y_0] = 2 + 0.1\ (1 + 0 \times 2) = 2.1$

By modified Euler's formula

$y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1)] = y_0 + \frac{h}{2}[1 + x_0y_0 + 1 + x_1y_1]$

$= 2 + \frac{0.1}{2}[1 + 0 + 1 + 0.1 \times 2.1] = 2.1105$

$y_1^{(2)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(1)})] = y_0 + \frac{h}{2}[1 + x_0y_0 + 1 + x_1 y_1^{(1)}]$

$= 2 + \frac{0.1}{2}[1 + 0 + 1 + 0.1 \times 2.1105] = 2.11055$

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$$y_1^{(3)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(2)})] = y_0 + \frac{h}{2}[1 + x_0y_0 + 1 + x_1y_1^{(2)}]$$

$$= 2 + \frac{0.1}{2}[1 + 0 + 1 + 0.1 \times 2.11055] = 2.11055$$

Now we find that $y_1^{(2)} = y_1^{(3)}$. So we stop here. Hence the required value $y(0.1) = 2.11055$.

Now, starting value of

$$y_2 = y_1 + hf(x_0 + h, y_1) = y_1 + h[1 + (x_0 + h)y_1]$$

$$= 2.11055 + 0.1[1 + (0 + 0.1) \times 2.11055] = 2.23166$$

$$y_2^{(1)} = y_1 + \frac{h}{2}[f(x_0 + h, y_1) + f(x_0 + 2h, y_2)]$$

$$= y_1 + \frac{h}{2}[1 + x_0 + h)y_1 + 1 + (x_0 + 2h)y_2]$$

$$= 2.11055 + \frac{0.1}{2}[1 + (0 + 0.1) \times 2.11055 + 1 + (0 + 2 \times 0.1) \times 2.23266]$$

$$= 2.24342$$

$$y_2^{(2)} = y_1 + \frac{h}{2}[f(x_0 + y, y_1) + f(x_0 + 2h, y_2^{(1)})]$$

$$= y_1 + \frac{h}{2}[1 + (x_0 + h)y_1 + 1 + (x_0 + 2h)\, y_2^{(1)}]$$

$$= 2.11055 + \frac{0.1}{2}[1 + (0 + 0.1 \times 2.11055 + 1 + (0 + 2 \times 0.1) \times 2.24342]$$

$$= 2.24354$$

$$y_2^{(3)} = y_1 + \frac{h}{2}[f(x_0 + h, y_1) + f(x_0 + 2h, y_2^{(2)}]$$

$$= y_1 + \frac{h}{2}[1 + (x_0 + h)y_1 + 1 + (x_0 + 2h)\, y_2^{(2)}]$$

$$= 2.11055 + \frac{0.1}{2}[1 + (0 + 0.1) \times 2.11055 + 1 + (0 + 2 \times 0.1) \times 2.24354]$$

$$= 2.24354$$

$\therefore$ we find that $y_2^{(2)} = y_2^{(3)}$. So we stop here. Hence the required value of $y(0.2)$ is 2.24354.

Now staring value of

$$y_3 = y_2 + hf(x_0 + 2h, y_2) = y_2 + h[1 + (x_0 + 2h)y_2]$$

$$= 2.24354 + 0.1 \times [1 + (0 + 2 \times 0.1) \times 2.24354] = 2.38841$$

$$y_3^{(1)} = y_2 + \frac{h}{2}[f(x_0 + 2h, y_2) + f(x_0 + 3h, y_3)]$$

$$= y_2 + \frac{h}{2}[1 + (x_0 + 2h)y_2 + 1 + x_0 + 3h)y_3]$$

$$= 2.24354 + \frac{0.1}{2}[1 + (0 + 2 \times 0.1) \times 2.24354 + 1$$

$$+ (0 + 3 \times 0.1) \times 2.38841]$$

$$= 2.4018$$

$$y_3^{(2)} = y_2 + \frac{h}{2}[f(x_0 + 2h, y_2) + f(x_0 + 3h, y_3^{(1)}]$$

$$= y_2 + \frac{h}{2}[1 + (x_0 + 2h)y_2 + 1 + (x_0 + 3h)\, y_3^{(1)}]$$

$$= 2.24354 + \frac{0.1}{2}[1 + (0 + 2 \times 0.1) \times 2.24354 +$$

$$+ (0 + 3 \times 0.1) \times 2.4018] = 2.4020$$

$$y_3^{(3)} = y_2 + \frac{h}{2}[f(x_0 + 2h, y_2) + f(x_0 + 3h, y_3^{(2)}]$$

$$= y_2 + \frac{h}{2}[1 + (x_0 + 2h)y_2 + 1 + (x_0 + 3h)y_3^{(2)}]$$

$$= 2.24354 + \frac{0.1}{2}[1 + (0 + 2 \times 0.1) \times 2.24354 + 1$$

$$+ (0 + 3 \times 0.1) \times 2.4018] = 2.4020$$

we find that $y_3^{(2)} = y_3^{(3)}$. So we stop here.

Hence the required value of $y(0.3) = 2.4020$

Example–31. Using modified Euler's method solve $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2}$, $y(1) = 1$. Evalute $y(1.3)$.

Solution : Given $\frac{dy}{dx} = \frac{1}{x^2} - \frac{y}{x} = \frac{1 - xy}{x^2}$ with $y(1) = 1$

Here $f(x, y) = \frac{1 - xy}{x^2}$, $x_0 = 1$, $y_0 = 1$

Let us take $h = 0.1$, By Euler's formula, we have

$$y_1 = y_0 + hf(x_0, y_0) = y_0 + h\left[\frac{1 - x_0 y_0}{x_0^2}\right] = 1 + 0.1 \times 0 = 1$$

By modified Euler's formula

$$y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1)] = y_0 + \frac{h}{2}\left[\frac{1 - x_0y_0}{x^2} + \frac{1 - x_1 y_1}{x_1^2}\right]$$

$$= 1 + \frac{0.1}{2}\left[\frac{1 - 1}{1} + \frac{1 - 1.1 \times 1}{11^2}\right] = 0.99587$$

$$y_1^{(2)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_0 + h, y_1^{(0)}]$$

$$= y_0 + \frac{h}{2}\left[\frac{1 - x_0y_0}{x_0^2} + \frac{1 - (x_0 + h)y_1^{(1)}}{(x_0 + h)^2}\right]$$

$$= 1 + \frac{0.1}{2}\left[0 + \frac{1 - 1.1 \times 0.99587}{(1.1)^2}\right] = 0.99606$$

$$y_1^{(3)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_0 + h, y_1^{(2)})]$$

$$= y_0 + \frac{h}{2}\left[\frac{1 - x_0y_0}{x_0^2} + \frac{1 - (x_0 + h)y_1^{(2)}}{(x_0 + h)^2}\right]$$

$$= 1 + \frac{0.1}{2}\left[0 + \frac{1 - 1.1 \times 0.99606}{(1.1)^2}\right] = 0.99605$$

$$y_1^{(4)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_0 + h, y_1^{(3)}]$$

$$= 1 + \frac{0.1}{2}\left[0 + \frac{0.1}{2} \times \frac{1 - 1.1 \times 0.99605}{(1.1)^2}\right] = 0.99605$$

$\therefore y_1^{(3)} = y_1^{(4)}$

Hence $y(1.1) = y_1 = 0.99605$

Now, $y_2 = y_1 + hf(x_0 + h, y_1) = y_1 + h\left[\frac{1 - (x_0 + h)y_1}{x_0^2}\right]$

$$= 0.99605 + 0.1\left[\frac{1 - 1.1 \times 0.99605}{(1.1)^2}\right] = 0.98814$$

$$y_2^{(1)} = y_1 + \frac{h}{2}[f(x_0 + h, y_1) + f(x_0 + 2h, y_2)]$$

$$= 0.99605 + \frac{0.1}{2}\left[\frac{1 - 1.1 \times 0.99605}{1.1^2} + \frac{1 - 1.2 \times 0.98814}{1.2^2}\right]$$

$$y_2^{(2)} = y_1 + \frac{h}{2}[f(x_0 + h, y_1) + f(x_0 + 2h, y_2^{(1)}] = 0.9857$$

$$y_2^{(3)} = y_1 + \frac{h}{2}[f(x_0 + h, y_1) + f(x_0 + 2h, y_2^{(2)})] = 0.9857$$

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$\therefore$ Final value of $y_2 = 0.9857$

$$y_3 = y_2 + hf(x_0 + 2h, y_2) = y_2 + h.\frac{1 - (x_0 + 2h)y_2}{(x_0 + 2h)^2} = 0.9730.$$

$$y_3^{(1)} = y_2 + \frac{h}{2}[f(x_0 + 2h, y_2) + f(x_0 + 3h, y_3)]$$

$$= y_2 + \frac{h}{2}\left[\frac{1 - (x_0 + 2h)y_2}{(x_0 + 2y)^2} + \frac{1 - (x_0 + 3h)y_3}{(x_0 + 2h)^2}\right] = 0.971$$

Continuing this process, we get

$y_3^{(2)} = 0.9716$, $y_3^{(3)} = 0.9662$, $y_3^{(4)} = 0.9662$

$\therefore y_3^{(3)} = y_3^{(4)}$, $\therefore$ Final value $y_3 = 0.9662$

Hence $y(1.3) = 0.9662$.

Example–32. Using modified Euler's method find the value of y when $x = 0.1$ of the equation $y' = x^2 + y$ when $y(0) = 1$.

[NUH–1999]

Solution : Given $y' = x^2 + y$ with $y(0) = 1$

Here $f(x, y) = x^2 + y$, $x_0 = 0$, $y_0 = 1$ and $h = 0.1$

$\therefore f(x_0, y_0) = x_0^2 + y_0 = 0 + 1 = 1$

$\therefore y_1^{(0)} = y_0 + hf(x_0, y_0) = 1 + 0.1 \times 1 = 1 + 0.1 = 1.1$

By Euler's Modified Formula for first approximation, we get

$$y_1^{(n+1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(n)})]$$

For $n = 0$, $y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(0)})]$

$$= 1 + \frac{0.1}{2}[(x_0^2 + y_0) + (x_1^2 + y_1^{(0)})]$$

$$= 1 + \frac{0.1}{2}[(0 + 1) + (0.1^2 + 1.1)] = 1.10550$$

For $n = 1$, $y_1^{(2)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(1)})]$

$$= 1 + \frac{0.1}{2}[(x_0^2 + y_0) + (x_1^2 + y_1^{(1)})]$$

$$= 1 + \frac{0.1}{2}[(0 + 1) + (0.1^2 + 1.10550)] = 1.10578$$

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$\therefore$ Final value of $y_2 = 0.9857$

$$y_3 = y_2 + hf(x_0 + 2h, y_2) = y_2 + h.\frac{1 - (x_0 + 2h)y_2}{(x_0 + 2h)^2} = 0.9730.$$

$$y_3^{(1)} = y_2 + \frac{h}{2}[f(x_0 + 2h, y_2) + f(x_0 + 3h, y_3)]$$

$$= y_2 + \frac{h}{2}\left[\frac{1 - (x_0 + 2h)y_2}{(x_0 + 2y)^2} + \frac{1 - (x_0 + 3h)y_3}{(x_0 + 2h)^2}\right] = 0.971$$

Continuing this process, we get

$y_3^{(2)} = 0.9716$, $y_3^{(3)} = 0.9662$, $y_3^{(4)} = 0.9662$

$\therefore y_3^{(3)} = y_3^{(4)}$, $\therefore$ Final value $y_3 = 0.9662$

Hence $y(1.3) = 0.9662$.

Example–32. Using modified Euler's method find the value of y when $x = 0.1$ of the equation $y' = x^2 + y$ when $y(0) = 1$.

[NUH–1999]

Solution : Given $y' = x^2 + y$ with $y(0) = 1$

Here $f(x, y) = x^2 + y$, $x_0 = 0$, $y_0 = 1$ and $h = 0.1$

$\therefore f(x_0, y_0) = x_0^2 + y_0 = 0 + 1 = 1$

$\therefore y_1^{(0)} = y_0 + hf(x_0, y_0) = 1 + 0.1 \times 1 = 1 + 0.1 = 1.1$

By Euler's Modified Formula for first approximation, we get

$$y_1^{(n+1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(n)})]$$

For $n = 0$, $y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(0)})]$

$$= 1 + \frac{0.1}{2}[(x_0^2 + y_0) + (x_1^2 + y_1^{(0)})]$$

$$= 1 + \frac{0.1}{2}[(0 + 1) + (0.1^2 + 1.1)] = 1.10550$$

For $n = 1$, $y_1^{(2)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(1)})]$

$$= 1 + \frac{0.1}{2}[(x_0^2 + y_0) + (x_1^2 + y_1^{(1)})]$$

$$= 1 + \frac{0.1}{2}[(0 + 1) + (0.1^2 + 1.10550)] = 1.10578$$

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For $n = 2$, $y_1^{(3)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(2)})]$

$= 1 + \frac{0.1}{2}[(x_0^2 + y_0) + (x_1^2 + y_1^{(2)})]$

$= 1 + \frac{0.1}{2}[(0 + 1) + (0.1^2 + 1.10578)] = 1.10579$

For $n = 3$, $y_1^{(4)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(3)})]$

$= 1 + \frac{0.1}{2}[(0 + 1) + (0.1^2 + 1.10579)] = 1.10579$

Since $y_1^{(3)} = y_1^{(4)} = 1.10579$, So, $y(0.1) = 1.10579$

Example–33. Using modified Euler's method find the val of $y(2)$ with $h = 0.1$, given that $\frac{dy}{dx} - \sqrt{xy} = 2$, $y(1) = 1$ [NUH-200

Solution : Given $y' = 2 + \sqrt{xy}$ with $y(1) = 1$

Here $f(x, y) = 2 + \sqrt{xy}$, $x_0 = 1$, $y_0 = 1$, $h = 0.1$

$\therefore f(x_0, y_0) = 2 + \sqrt{x_0 y_0}$

$\therefore y_1^{(0)} = y_0 + hf(x_0, y_0) = y_0 + h(2 + \sqrt{x_0 y_0})$

$= 1 + 0.1(2 + \sqrt{1 \times 1}) = 1 + .3 = 1.3$

By Euler's Modified formula for first approximation, we get

$y_1^{(n+1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(n)})] = y_0 + \frac{h}{2}[2 + \sqrt{x_0 y_0} + 2 + \sqrt{x_1 y_1}$

$= y_0 + \frac{h}{2}[4 + \sqrt{1 \times 1} + \sqrt{x_1 y_1^{(n)}}] = 1 + \frac{0.1}{2}[5 + \sqrt{1.1 \times y_1^{(n)}}]$

For $n = 0$, $y_1^{(1)} = 1 + \frac{0.1}{2}[5 + \sqrt{1.1 \times y_1^{(0)}}] = 1.30979$

For $n = 2$, $y_1^{(2)} = 1 + \frac{0.1}{2}[5 + \sqrt{1.1 \times y_1^{(1)}}] = 1.31002$

For $n = 3$, $y_1^{(3)} = 1 + \frac{0.1}{2}[5 + \sqrt{1.1 \times y_1^{(2)}}] = 1.31002$

It is clear that $y_1^{(2)} = y_1^{(3)} = 1.31002$, $\therefore y(1.1) = 1.31002$

For Second approximation, we get

$y_2^{(0)} = y_1 + hf(x_1, y_1) = y_1 + h(2 + \sqrt{x_1 y_1})$

$= 1.31002 + 0.1(2 + \sqrt{1.1 \times 1.31002} = 1.63006.$